

## The problem

Epigenomic measurements often suffer from noise, batch effects and irreproducibility. Imputation methods operate on processed signal, and assume that processing removes all batch effects. In the ENCODE Imputation Challenge, due to subtle train-test differences, a simple baseline outperformed most entrants [1].

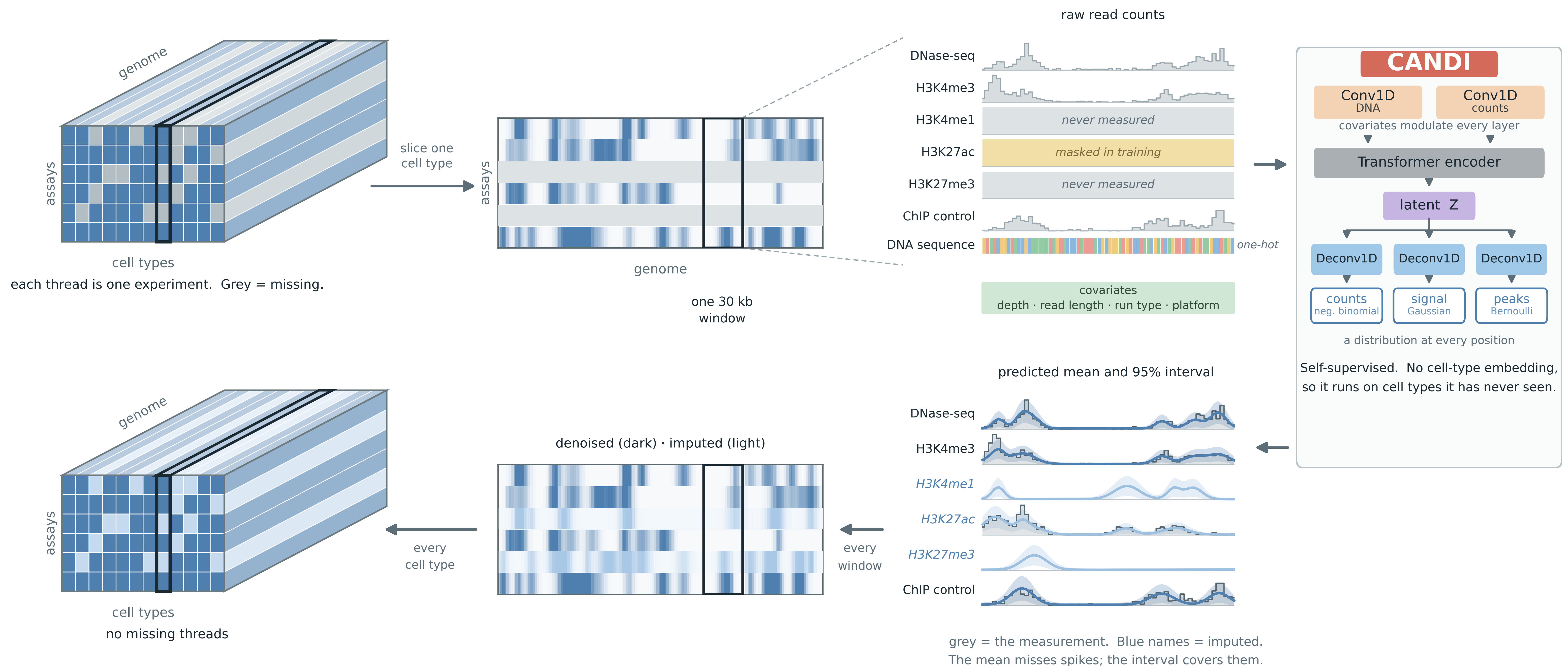
## CANDI

**Confidence-Aware Neural Denoising Imputer** [2] is a self-supervised deep learning method for epigenome imputation and denoising. It takes raw read counts, DNA sequence and experimental covariates, and outputs probability distributions over counts, signal and peaks. Conditioning on metadata rather than cell-type embeddings enables zero-shot generalization to unseen cell types.

## Key findings

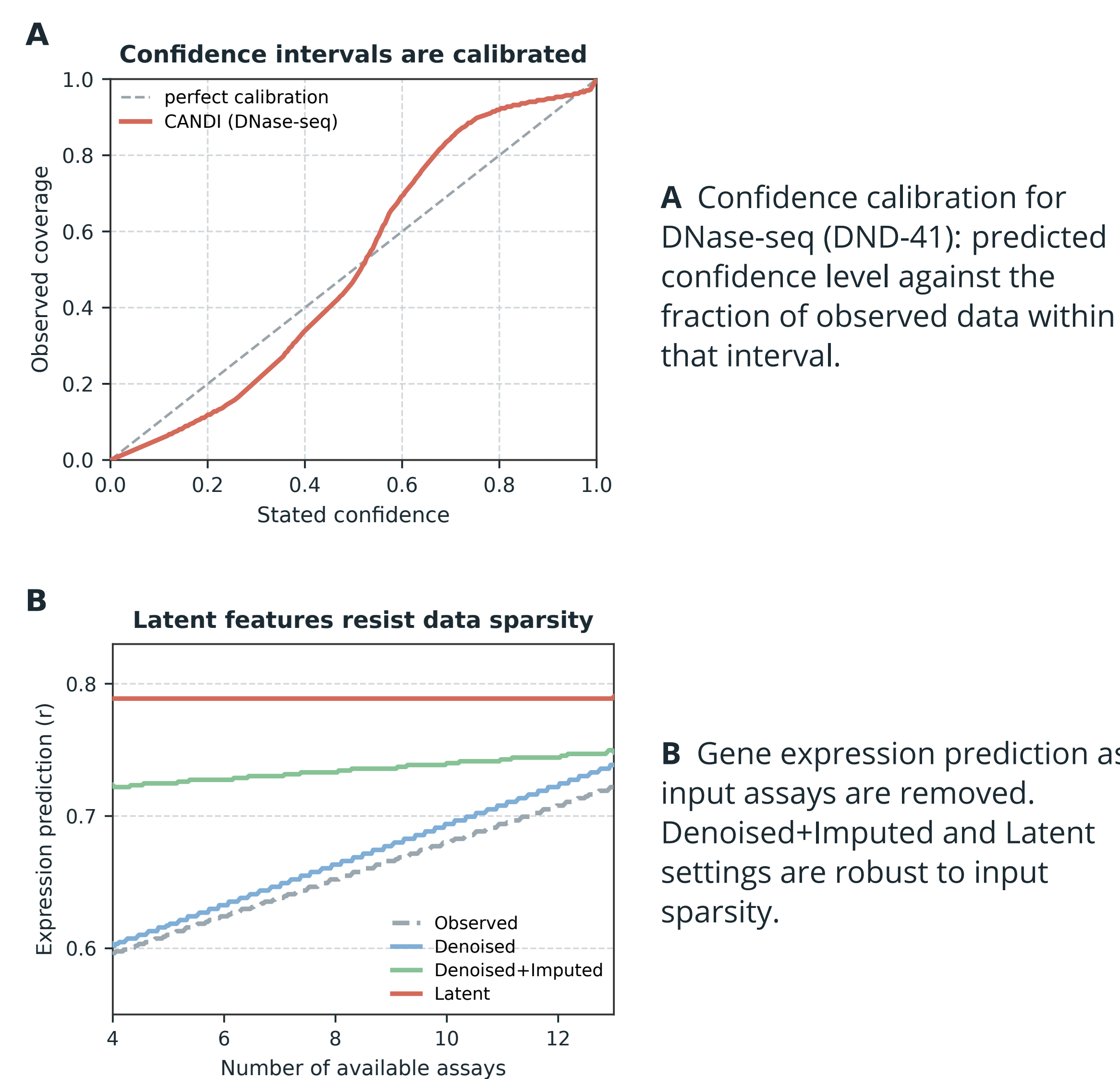
- CANDI ranks **1st of 26** on the ENCODE Imputation Challenge.
- CANDI provides calibrated uncertainty estimates.
- CANDI's imputed signals and latent space predict gene expression.

## CANDI schematic workflow



**Top:** for each 30 kb locus, CANDI takes observed read counts with the ChIP-seq control, four covariates per assay and the DNA sequence; grey assays were never measured, yellow ones are masked. **Bottom:** each output is a probability distribution, here a mean and 95% interval per assay; window by window, they fill every missing thread of the tensor.

## Downstream utility of CANDI

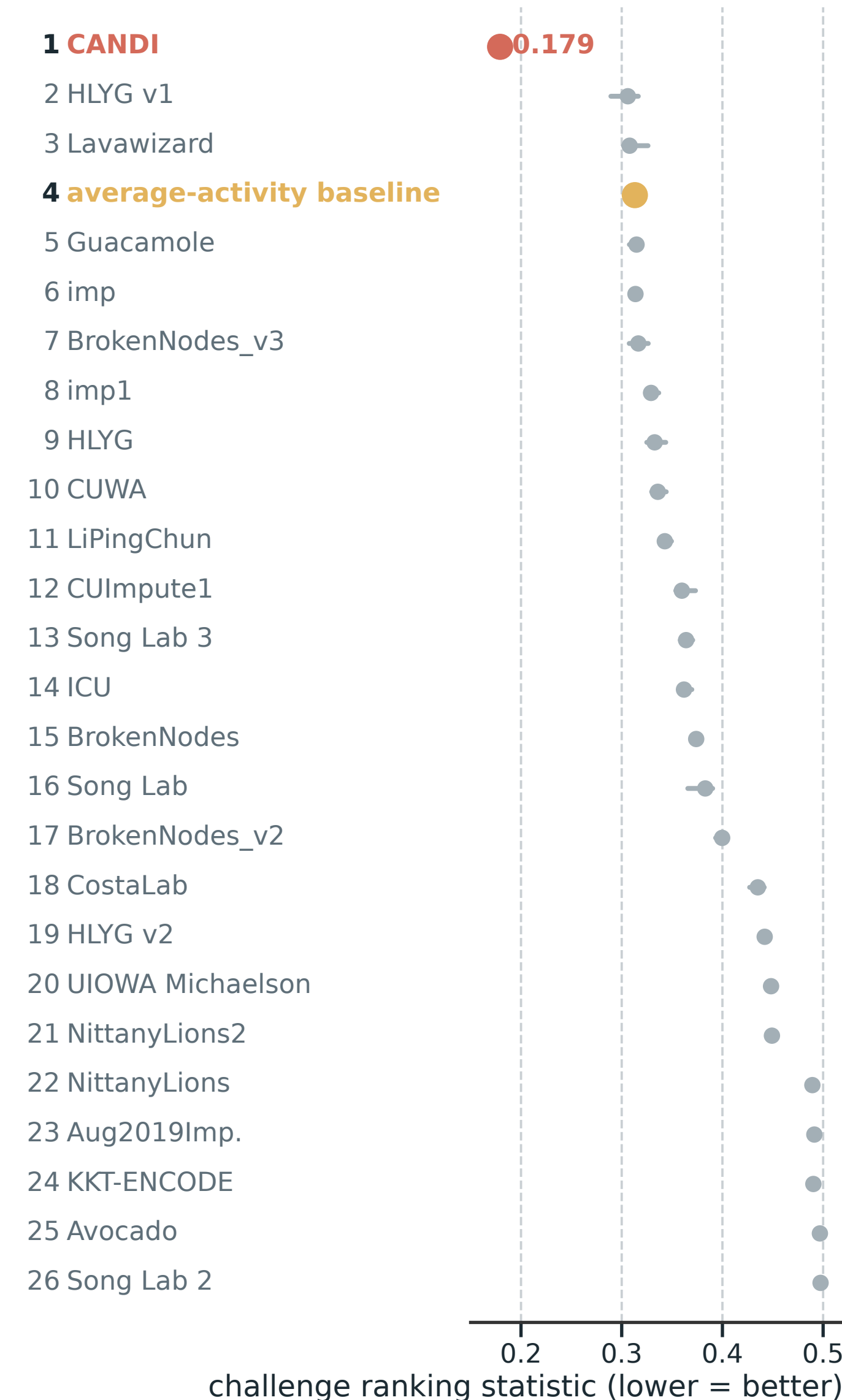


[1] Schreiber J, Boix C, *et al.* The ENCODE Imputation Challenge: a critical assessment of methods for cross-cell type imputation of epigenomic profiles. *Genome Biol* 24:79 (2023).

[2] Foroozandeh M, Diab AR, Libbrecht M. CANDI: self-supervised, confidence-aware denoising imputation of genomic data. *bioRxiv* 2025.01.23.634626 (2025).

Panels C–D: blind-test scoring from the Libbrecht lab's EIC reproduction (experiments 017, 025), which reproduces every published entry to a median relative difference of  $9 \times 10^{-10}$ ; *baseline* is the challenge's average-activity track. The correction was chosen after the challenge results were public; none of it is fitted on blind-test data.

## Fixing the signal-level shift in the ENCODE Imputation Challenge blind test



**C The leaderboard with CANDI added:** CANDI is first at 0.179 (bootstrap range 0.177–0.184), the next entry 0.306.

**The shift.** We followed the EIC train-test split: 51 blind-test experiments (12 cell types, 8 assays). The train and test sets differ subtly: every blind-test experiment is paired-end, against 15% of training experiments, and blind-test tracks sit at an assay-specific signal level. CANDI recovers the rank-ordering and structure of the signal but not its absolute magnitude; its raw output ranks 14th.

**The fix.** Per assay, CANDI's output passes through one layer,

$$\max(\beta_0 + \beta_1 \cdot \text{CANDI} + \beta_2 \cdot \text{baseline}, 0),$$

fitted by least squares on the 133 training experiments only. CANDI then ranks 1st.



**D** Eight of the nine measures (top-1%-imputed MSE not shown) relative to the baseline, median over 51 experiments. CANDI is first on three and top 8